Chapter 6 · Semester 1

GRAVITATION

Notes by Cikgu Suhaili

29 worked examples 2 simulations Exercises with answers

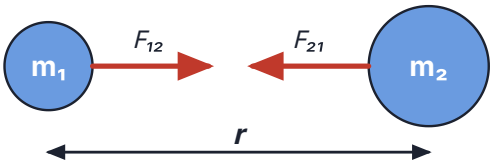
6.1 Newton's law of universal gravitation

Newton's law of universal gravitation

Every particle in the universe attracts every other particle with a force that is **directly proportional to the product of their masses** and **inversely proportional to the square of the distance between their centres**. The force acts along the line joining the two particles.

$$F = \frac{Gm_1m_2}{r^2}$$

G = universal gravitational constant = 6.67×10^{-11} N m^2 kg^{-2}



- F_{12} = force on object 1 by object 2; F_{21} = force on object 2 by object 1; $|F_{12}| = |F_{21}|$ (an action–reaction pair), even when the masses are different.
- The gravitational force is always **attractive** and is a **conservative** force.
- The weight of an object on Earth is the gravitational force between the object and the Earth.
- The gravitational force between the Earth and the Sun provides the centripetal force for the Earth's orbit.

Example 6.1.1

Calculate the gravitational force between two bodies of mass 3.0 kg and 2.0 kg whose centres are 50 cm apart.

Solution

$$F = \frac{(6.67 \times 10^{-11})(3.0)(2.0)}{(0.50)^2} = \mathbf{1.60 \times 10^{-9}\ N}$$

Example 6.1.2

Two objects attract each other with a force of 2.5×10^{-10} N when they are 0.25 m apart. Their total mass is 4.0 kg. Find their individual masses.

Solution

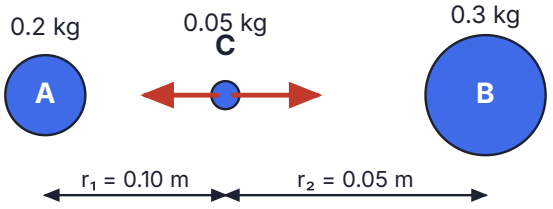
$m_1 m_2 = \frac{Fr^2}{G} = \frac{(2.5 \times 10^{-10})(0.25)^2}{6.67 \times 10^{-11}} = 0.234$; and $m_1 + m_2 = 4.0$

$m_1(4.0 - m_1) = 0.234 \Rightarrow m_1^2 - 4.0m_1 + 0.234 = 0$

$m_1 = 3.9$ kg, $m_2 = 0.1$ kg

Example 6.1.3

Particles A (0.2 kg) and B (0.3 kg) are 0.15 m apart. A third particle C (0.050 kg) is placed between them on the line joining them. Calculate (a) the gravitational force on C if it is 0.05 m from B, (b) the point on the line where there is no resultant gravitational force on C.



Solution

(a) $F_{CA} = G\frac{(0.05)(0.2)}{(0.10)^2} = 6.7 \times 10^{-11}$ N; $F_{CB} = G\frac{(0.05)(0.3)}{(0.05)^2} = 40.0 \times 10^{-11}$ N

Net force = $F_{CB} - F_{CA}$ = **33.3×10^{-11} N towards B**

(b) Let C be x from A: $G\frac{m_C m_A}{x^2} = G\frac{m_C m_B}{(0.15 - x)^2} \Rightarrow \frac{0.15 - x}{x} = \sqrt{\frac{0.3}{0.2}}$

$x = 0.067$ m from A

Exercises: Newton's law of gravitation

1. One small ball has mass 0.00150 kg and another 0.870 kg. Their centres are 0.100 m apart. Find the gravitational force each exerts on the other.

   Answer

   8.70×10^{-12} N

2. Three particles lie on a straight line, far from other objects: A (363 kg), B (517 kg) 0.500 m to the right of A, and C (154 kg) 0.250 m to the right of B. Find the net gravitational force on (a) A, (b) B, (c) C.

   Answer

   (a) 5.67×10^{-5} N to the right (b) 3.49×10^{-5} N to the right (c) 9.16×10^{-5} N to the left

3. The mass of the Earth is 6.0×10^{24} kg and of the Moon 7.4×10^{22} kg. Their centres are 3.8×10^{8} m apart. Find the point on the line between them where the net gravitational force is zero.

   Answer

   3.42×10^{8} m from the centre of the Earth

6.2 Gravitational field

Gravitational field

is a **region in which an object with mass experiences a force**.

Gravitational field strength, g

is the **gravitational force per unit mass** acting on a small mass (test mass) placed at that point in the field.

$$g = \frac{F}{m} = \frac{GM}{r^2}$$

- Unit: N kg^{-1} (equivalent to m s^{-2}).
- g is a vector, directed towards the centre of the mass M.
- g is also the **acceleration due to gravity**.

Gravitational force on mass m at distance r from a planet of mass M: $F = \frac{GMm}{r^2}$ (1)

If released, it falls towards the planet with acceleration a_g: $F = ma_g$ (2)

(1) = (2): $a_g = \frac{GM}{r^2}$. At the Earth's surface ($r = R_E$): $g = \frac{GM_E}{R_E^2}$

Equivalently, weight $mg = \frac{GM_E m}{R_E^2}$, so $GM_E = gR_E^2$

Example 6.2.1

Taking the distance from a baseball to the centre of the Earth as the Earth's radius, 6.38×10^6 m, find the mass of the Earth. ($g = 9.81\ \text{m s}^{-2}$)

Solution

$$M_E = \frac{gR_E^2}{G} = \frac{(9.81)(6.38 \times 10^6)^2}{6.67 \times 10^{-11}} = \mathbf{5.99 \times 10^{24}\ kg}$$

Example 6.2.2

On a distant planet the acceleration due to gravity is $5.00\ \text{m s}^{-2}$ and the radius is about 4560 km. Estimate the mass of the planet.

Solution

$$M = \frac{gr^2}{G} = \frac{(5.00)(4.56 \times 10^6)^2}{6.67 \times 10^{-11}} = \mathbf{1.56 \times 10^{24}\ kg}$$

Example 6.2.3

The Earth is a uniform sphere of radius 6.4×10^6 m and mean density $5500\ \text{kg m}^{-3}$. Calculate the acceleration due to gravity at its surface.

Solution

$M = \rho(\frac{4}{3}\pi R^3)$, so $g = \frac{GM}{R^2} = \frac{4}{3}\pi G\rho R$

$g = \frac{4}{3}\pi(6.67 \times 10^{-11})(5500)(6.4 \times 10^6) = \mathbf{9.83\ m\ s^{-2}}$

Example 6.2.4

An object weighs 100 N on the Earth's surface. What does it weigh on the Moon's surface? (r_m = $0.25r_E$, M_m = $0.01M_E$)

Solution

$$W_E = \frac{GM_E m}{r_E^2} = 100 \text{ N}$$

$$W_m = \frac{G(0.01M_E)m}{(0.25r_E)^2} = \frac{GM_E m}{r_E^2}\left(\frac{0.01}{(0.25)^2}\right) = 100 \times 0.16 = \mathbf{16\ N}$$

Exercises: field strength

1. The mass of the Earth is about 81 times the mass of the Moon and its radius is four times the Moon's. Find the acceleration due to gravity on the Moon. (g_E = 9.81 m s^{-2})

   Answer

   1.94 m s^{-2}

2. A planet has 1.5 times the radius of the Earth but the same mass. What is the acceleration due to gravity at its surface?

   Answer

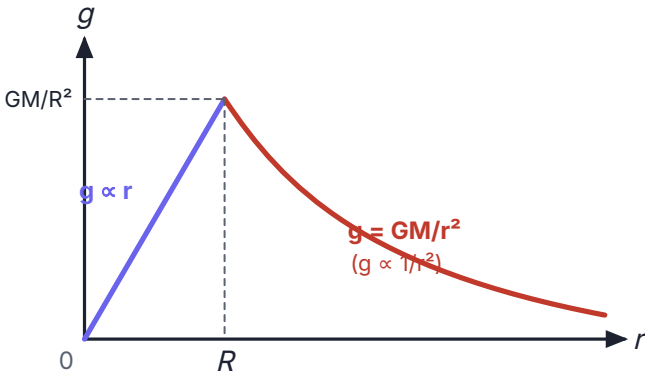
   4.36 m s^{-2}

Variation of g with distance from the centre of the Earth

On the surface ($r = R_E$): $g = \dfrac{GM_E}{R_E^2}$, constant if the Earth's radius is uniform.

Above the surface ($r > R_E$, $r = R_E + h$): $g = \dfrac{GM}{r^2}$, so $g \propto \dfrac{1}{r^2}$; $g \to 0$ as $r \to \infty$.

Below the surface ($r < R_E$), uniform density: only the mass inside radius r counts, $M = \rho(\frac{4}{3}\pi r^3)$, so $g = G\rho(\frac{4}{3}\pi r)$: $g \propto r$. g decreases linearly with depth.



Example 6.2.5

The Earth is a uniform sphere of radius 6.4×10^6 m and mass 6.0×10^{24} kg. Calculate g (i) at the surface, (ii) at a height of 0.5 times its radius above the surface.

Solution

(i) $g = \dfrac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{(6.4 \times 10^6)^2} = \mathbf{9.8\ N\ kg^{-1}}$

(ii) $r_1 = r + 0.5r = 1.5r$; $\dfrac{g_1}{g} = (\dfrac{r}{1.5r})^2 = 0.444 \Rightarrow g_1 = 0.444 \times 9.8 = \mathbf{4.35\ N\ kg^{-1}}$

Variation of g with latitude

- The Earth spins with angular velocity $\omega = \dfrac{2\pi}{T} = 7.27 \times 10^{-5}$ rad s^{-1} (T = 24 h).
- A body at latitude λ moves in a circle of radius $R\cos\lambda$, so part of the gravitational pull provides the centripetal force: $g_{eff} = \dfrac{GM}{R^2} - \omega^2 R\cos^2\lambda$.
- At the **equator** ($\lambda = 0°$) g is a minimum; at the **poles** ($\lambda = 90°$, $\cos\lambda = 0$) g is a maximum.
- The Earth is also slightly flattened (its radius is larger at the equator), which makes g larger at the poles as well.

STPM: calculations of the variation with latitude are limited to the equator and the poles.

Example · g at the equator and at the poles

Given $R = 6.37 \times 10^6$ m, $\omega = 7.27 \times 10^{-5}$ rad s^{-1} and $\frac{GM}{R^2} = 9.83$ N kg^{-1}, find g at the poles and at the equator.

Solution

$\omega^2 R = (7.27 \times 10^{-5})^2 (6.37 \times 10^6) = 0.0337$ m s^{-2}

Poles: $g = 9.83$ N kg^{-1}; equator: $g = 9.83 - 0.034 = 9.80$ N kg^{-1} (a difference of about 0.3%)

6.3 Gravitational potential

Gravitational potential energy, U

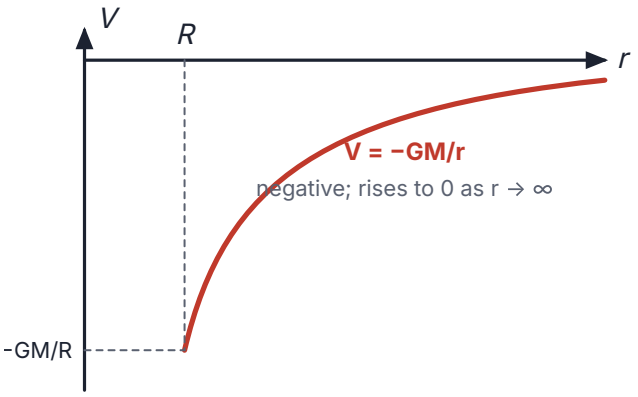
of a mass m at a point is the **work done by an external force in bringing the mass from infinity to that point**.

Bringing m from infinity to r, the external force balances gravity: $W = \int_\infty^r \frac{GM_E m}{r^2}\,dr$ (taken towards the Earth)

$$W = -\frac{GM_E m}{r} = U$$

$$U = -\frac{GMm}{r}$$

Unit: J



- At the Earth's surface, $U = -\frac{GM_E m}{R_E}$.
- The **minus sign** shows that the potential energy at the surface is less than at infinity (where $U = 0$). A body at infinity would "fall" towards the Earth; a body on the Earth does not "fall" to infinity.

Gravitational potential, V

at a point is the **gravitational potential energy per unit mass** of a small test mass placed at that point; equivalently, the work done per unit mass by an external force in bringing a small mass from infinity to that point.

$$V = \frac{U}{m} = -\frac{GM}{r}$$

Unit: J kg^{-1}

Variation of V with distance from the Earth

- For $r \geq R_E$, $V = -GM/r$: always negative, and it increases (becomes less negative) as r increases, with $V \to 0$ as $r \to \infty$.
- V is most negative at the surface: $V_{surface} = -\frac{GM}{R_E} = -gR_E \approx -6.3 \times 10^7$ J kg^{-1}.
- $V \propto 1/r$ while $g \propto 1/r^2$, so V falls off more slowly than g.

Example 6.3.1

Find the gravitational potential at the surface of the Earth, if the surface field strength is 9.8 N kg^{-1} and the mean radius is 6400 km.

Solution

$$g = \frac{GM_E}{R_E^2} \Rightarrow \frac{GM_E}{R_E} = gR_E; \quad V = -gR_E = -9.8(6.4 \times 10^6) = \mathbf{-6.3 \times 10^7\ J\ kg^{-1}}$$

Example 6.3.2

A 200 kg satellite orbits at an altitude of 2.00×10^6 m. Calculate its gravitational potential energy. ($M_E = 6.0 \times 10^{24}$ kg, $R_E = 6.4 \times 10^6$ m)

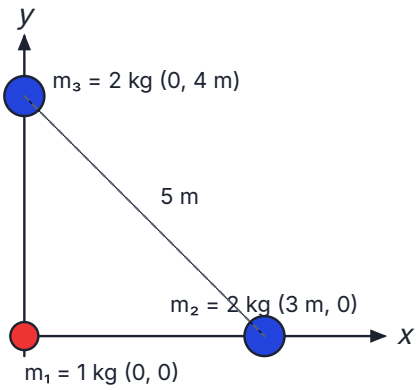
Solution

$r = R_E + h = 6.4 \times 10^6 + 2.00 \times 10^6 = 8.4 \times 10^6$ m

$$U = -\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(200)}{8.4 \times 10^6} = \mathbf{-9.53 \times 10^9\ J}$$

Example 6.3.3

Calculate the total gravitational potential energy of the system of three masses shown. (G = 6.7 × 10^{-11} N m^2 kg^{-2})



Solution

$$U = U_{12} + U_{13} + U_{23} = -\frac{Gm_1m_2}{r_{12}} - \frac{Gm_1m_3}{r_{13}} - \frac{Gm_2m_3}{r_{23}}$$

$$= -6.7 \times 10^{-11}\left[\frac{(1)(2)}{3} + \frac{(1)(2)}{4} + \frac{(2)(2)}{5}\right] = \mathbf{-1.3 \times 10^{-10}\ J}$$

Work done and the change in potential energy

Work done (by an external force) to move m from r_1 to r_2 = change in gravitational potential energy: $W = U_2 - U_1 = \Delta U$

$$W = -\frac{GMm}{r_2} - \left(-\frac{GMm}{r_1}\right) = m(V_2 - V_1)$$

$$W = \Delta U = m\Delta V$$

Example 6.3.4

The potential energy of an object at A, a distance R from a planet, is −900 J. Calculate (a) the potential energy at B, a distance $3R$, (b) ΔU from A to B, (c) ΔU from B to A, (d) the work done to move the object from A to B.

Solution

(a) $U \propto \frac{1}{r}$: $\frac{U_B}{U_A} = \frac{R}{3R} \Rightarrow U_B =$ **−300 J**

(b) $\Delta U_{AB} = U_B - U_A = (-300) - (-900) =$ **600 J**

(c) $\Delta U_{BA} = U_A - U_B =$ **−600 J**

(d) $W_{AB} = \Delta U_{AB} =$ **600 J**

Example 6.3.5

How much energy is needed to move a 1000 kg object from the Earth's surface to an altitude of twice the Earth's radius? ($M_E = 6.0 \times 10^{24}$ kg, $R_E = 6.4 \times 10^6$ m)

Solution

Altitude $2R_E$ means $r = 3R_E$ from the centre.

$W = m(V_f - V_i) = GMm(\frac{1}{R_E} - \frac{1}{3R_E}) = \frac{2GMm}{3R_E} =$ **4.17 × 10¹⁰ J**

ΔU over small distances: U = mgh is a special case

Move m from the surface ($r = R_E$) up a height h: $\Delta U = -\frac{GM_Em}{R_E + h} - (-\frac{GM_Em}{R_E})$

$= GM_Em(\frac{1}{R_E} - \frac{1}{R_E + h}) = GM_Em\,\frac{h}{R_E(R_E + h)} = GM_Em\,\frac{h}{R_E^2(1 + h/R_E)}$

If $h \ll R_E$, $h/R_E \approx 0$: $\Delta U = \frac{GM_E}{R_E^2}mh = \boldsymbol{mgh}$, where $g = \frac{GM_E}{R_E^2}$

Relationship between g and V

Move a 1 kg test mass a small distance $\mathrm{d}r$ away from M. The gain in potential equals the work done per unit mass against the field: $\mathrm{d}V = -g\,\mathrm{d}r$

$$g = -\frac{\mathrm{d}V}{\mathrm{d}r}$$

g is the negative of the gradient of the V–r graph (the potential gradient). Units: J kg^{-1} m^{-1} = N kg^{-1}.

- The minus sign shows that g acts in the direction of **decreasing** potential, i.e. towards M.
- Check: $V = -\frac{GM}{r} \Rightarrow \frac{\mathrm{d}V}{\mathrm{d}r} = \frac{GM}{r^2} \Rightarrow g = -\frac{GM}{r^2}$: magnitude GM/r^2, directed towards M.
- Reading g from a V–r graph: $g = -\frac{\Delta V}{\Delta r} = -\frac{8.5 \times 10^6}{1.0 \times 10^6} = -8.5$ N kg^{-1}, i.e. 8.5 N kg^{-1} towards the centre.

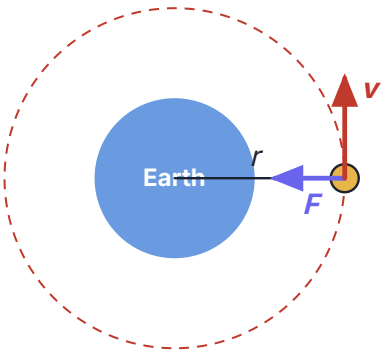
6.4 Satellite motion in a circular orbit

Satellite

is a body **orbiting a larger mass under the influence of its gravitational force**. Examples: the Moon (of the Earth), communication satellites, the Earth (of the Sun).

The gravitational attraction provides the centripetal force: $F_g = F_c$

$$\frac{GM_E m}{r^2} = \frac{mv^2}{r}$$



Orbital speed, $v = \sqrt{\frac{GM_E}{r}}$

- Orbital speed is the speed needed to stay in an orbit of fixed radius.
- It is **independent of the mass** of the satellite: a heavy satellite and a light one in the same orbit have the same speed.

Period: $T = \frac{2\pi r}{v}$. Substitute $v = \sqrt{\frac{GM_E}{r}}$:

$$T = \frac{2\pi r^{3/2}}{\sqrt{GM_E}} \Rightarrow T^2 = \frac{4\pi^2 r^3}{GM_E}$$

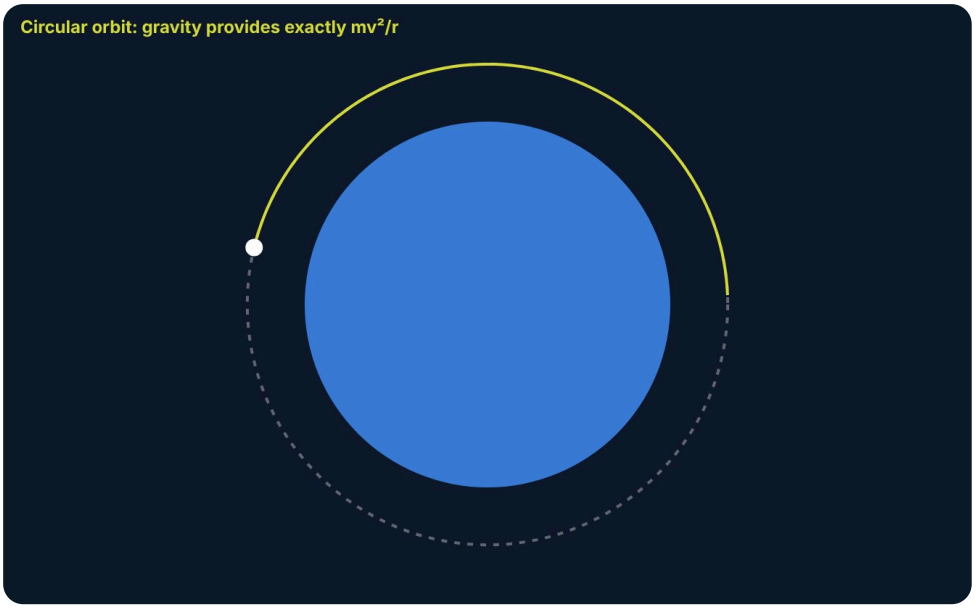
Since G and M_E are constant, $T^2 \propto r^3$ (Kepler's third law).

Geosynchronous satellites

- "Geosynchronous" means synchronised with the Earth's rotation.
- A **geosynchronous orbit** has the same rotational period (24 h) and direction as the rotation of the Earth.
- A geosynchronous satellite whose orbit lies over the equator is a **geostationary satellite**: it stays above the same point on the Earth, so it is used for communication.

Simulation **Satellite orbits: speed, period and escape**

Choose the orbit radius and launch speed. At exactly √(GM/r) the orbit is a circle; slower gives an ellipse that dips towards the Earth; at √2 times the circular speed the satellite escapes.



orbit radius r	circular speed √(GM/r)	period 2πr/v	escape speed √(2GM/r)
8.37 × 10⁶ m	**6.90 km/s**	**2.12 h**	**9.76 km/s**

launch speed	energy per kg now
6.90 km/s	**−23.8 MJ/kg**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 6 to use the simulation.

Example 6.4.1

A communication satellite orbits the Earth with a period of 24 hours. Find its height above the Earth's surface. (R_E = 6.4 × 10^6 m, M_E = 6.0 × 10^{24} kg, G = 6.67 × 10^{-11} N m^2 kg^{-2})

Solution

$T = 24 \times 60 \times 60 = 86\,400$ s

$r = \left(\frac{GM_E T^2}{4\pi^2}\right)^{1/3} = 4.23 \times 10^7$ m

$h = 4.23 \times 10^7 - 6.4 \times 10^6 =$ **3.59×10^7 m**

Example 6.4.2

The Moon takes 27.3 days to orbit the Earth. Calculate its average distance from the Earth. (g = 9.81 N kg^{-1}, R_E = 6400 km)

Solution

$GM_E = gR_E^2$; $T = 27.3 \times 24 \times 3600 = 2.36 \times 10^6$ s

$r^3 = \frac{gR_E^2 T^2}{4\pi^2} \Rightarrow$ **$r = 3.8 \times 10^8$ m**

Example 6.4.3

A satellite orbits 596 km above the Earth's surface. Find (a) its speed, (b) its period, (c) its speed and period if it moves to an orbit 400 km above the surface. (R_E = 6.38 × 10^6 m, M_E = 5.98 × 10^{24} kg)

Solution

(a) r = 6.38 × 10^6 + 596 × 10^3 = 6.98 × 10^6 m; $v = \sqrt{\frac{GM}{r}}$ = **7.56 × 10^3 m s^{-1}**

(b) $T = \frac{2\pi r}{v}$ = **5.80 × 10^3 s** (1.61 h)

(c) $v^2 r = GM$ = constant ⇒ $v_1^2 r_1 = v_2^2 r_2$, with r_2 = 6.78 × 10^6 m

(7.56 × 10^3)^2(6.98 × 10^6) = v_2^2(6.78 × 10^6) ⇒ **v_2 = 7.67 × 10^3 m s^{-1}**; $T_2 = \frac{2\pi r_2}{v_2}$ = **5554 s (1.54 h)**

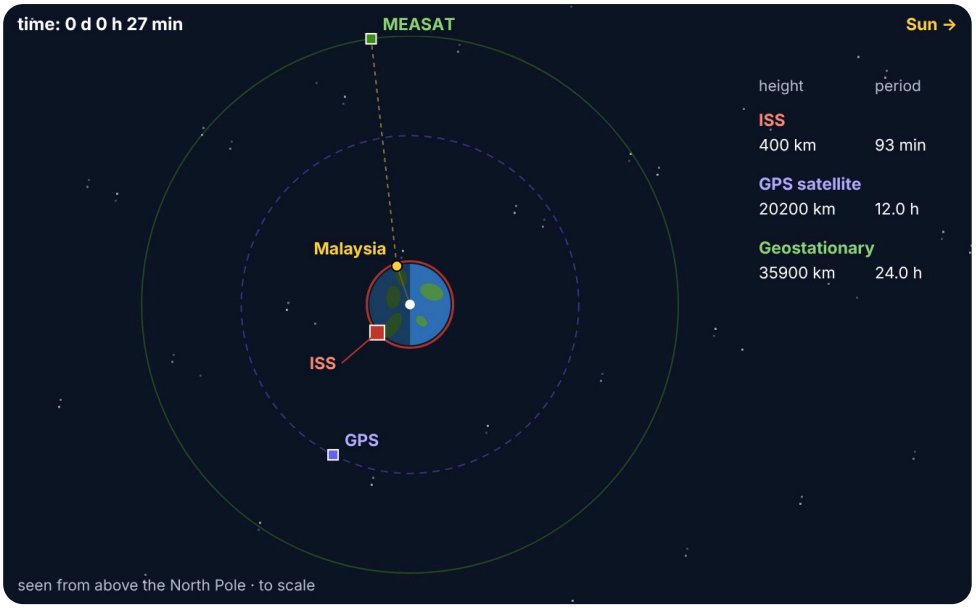
Real satellites above Malaysia

- **International Space Station (ISS):** a low orbit about 400 km up. It moves at almost 8 km s^{-1} and goes round the Earth in about 93 minutes, about 15 times a day.
- **GPS satellites:** about 20 200 km up, period 12 hours. The phone in your pocket uses signals from several of them to find your position.
- **Geostationary satellites** (e.g. the MEASAT satellites used for satellite TV in Malaysia): a geosynchronous orbit about 35 800 km above the **equator**, moving west to east. They stay above the same point on the Earth, so a satellite dish can be fixed pointing at one spot in the sky.
- The higher the orbit, the **slower** the satellite and the **longer** its period: $v = \sqrt{(GM/r)}$ and $T^2 \propto r^3$.

Simulation

The ISS, GPS and a geostationary satellite, drawn to scale

A view looking down on the North Pole, with the orbits drawn to the correct scale. Speed up time and watch: the ISS races round, a GPS satellite goes round twice a day, and the geostationary satellite stays above Malaysia as the Earth turns.



ISS: height	orbit radius r = R + h	speed v = √(GM/r)	period T = 2πr/v	orbits per day
400 km	**6800 km**	**7.67 km s^{-1}**	**92.8 min**	**15.5**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 6 to use the simulation.

Example 6.4.7

The International Space Station orbits 400 km above the Earth's surface. Find (a) its orbital speed, (b) its period in minutes, (c) the number of orbits it makes in one day. (R_E = 6.4 × 10^6 m, M_E = 6.0 × 10^{24} kg, G = 6.67 × 10^{-11} N m^2 kg^{-2})

Solution

$r = 6.4 \times 10^6 + 0.4 \times 10^6 = 6.8 \times 10^6$ m

(a) $v = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{6.8 \times 10^6}}$ = **7.67 × 10^3 m s^{-1}**

(b) $T = \frac{2\pi r}{v} = \frac{2\pi(6.8 \times 10^6)}{7.67 \times 10^3}$ = 5.57 × 10^3 s = **92.8 min**

(c) $\frac{24 \times 60}{92.8}$ = **15.5 orbits a day**

Exercises: orbits

1. The Earth orbits the Sun (mass 2 × 10^{30} kg) once a year. Find the distance of the Earth from the Sun.

   Answer

   1.50 × 10^{11} m

2. A satellite 36 000 km above the Earth's surface orbits once in 24 h. Calculate the mass of the Earth.

   Answer

   6.02 × 10^{24} kg

3. Satellites A and B are in different circular orbits about the Earth. A's orbital speed is twice B's. Find T_A/T_B.

   Answer

   0.125 ($r = GM/v^2$ and $T = 2\pi r/v = 2\pi GM/v^3$, so $T \propto 1/v^3$)

4. A satellite orbits an unknown planet at 1.70 × 10^4 m s^{-1} with radius 5.25 × 10^6 m. A second satellite orbits the same planet with radius 8.60 × 10^6 m. Find its orbital speed.

   Answer

   1.33 × 10^4 m s^{-1}

Energies of a satellite

Centripetal force = gravitational force: $\frac{mv^2}{r} = \frac{GM_E m}{r^2} \Rightarrow mv^2 = \frac{GM_E m}{r}$

K.E. = $\frac{1}{2}mv^2 = \frac{GM_E m}{2r}$; P.E. = $-\frac{GM_E m}{r}$

Total energy = K.E. + P.E. = $\frac{GM_E m}{2r} - \frac{GM_E m}{r}$

$$\text{K.E.} = \frac{GMm}{2r} \qquad \text{P.E.} = -\frac{GMm}{r} \qquad E = -\frac{GMm}{2r}$$

The total energy is negative: the satellite is bound to the Earth.

Example 6.4.4

An astronaut puts a bowling ball of mass 7.20 kg into a circular orbit 350 km above the Earth. Find (a) its K.E., (b) its P.E., (c) its total mechanical energy. (R_E = 6.4 × 10^6 m, M_E = 6 × 10^{24} kg)

Solution

r = 6.4 × 10^6 + 0.35 × 10^6 = 6.75 × 10^6 m

(a) K.E. = $\frac{GM_E m}{2r}$ = **2.13 × 10^8 J**

(b) P.E. = $-\frac{GM_E m}{r}$ = **−4.27 × 10^8 J**

(c) E = K.E. + P.E. = **−2.13 × 10^8 J**

Weightlessness

- We judge our weight from the upward push of the floor (the reading on a scale): this is the **apparent weight**, R. The **actual weight** is the gravitational force, $W = mg$.
- If our feet are completely unsupported (R = 0) we experience **weightlessness**.

Lift	Newton's second law	Apparent weight
at rest or uniform velocity	$R = mg$	apparent weight = actual weight
accelerating downward	$ma = mg - R$	$R = m(g - a)$: less; $R = 0$ when $a = g$ (free fall)
accelerating upward	$ma = R - mg$	$R = m(g + a)$: more

- An astronaut in an orbiting spacecraft moves with constant speed round the orbit. The only force on him is gravity, which provides the centripetal force: $F_c = F$, so his acceleration $a = g'$, the acceleration due to gravity at that height.
- He and the spacecraft fall freely together, so the walls exert no force on him ($R = 0$). He experiences **apparent weightlessness** and floats about in the spacecraft. Gravity has not disappeared.

Example 6.4.5

A boy of mass 70 kg is in a lift moving upward with acceleration 2 m s^{-2}. What is his apparent weight?

Solution

$$ma = R - mg \Rightarrow R = m(g + a) = 70(9.81 + 2) = \mathbf{826.7\ N}\ \text{(he feels heavier)}$$

Example 6.4.6

A 70 kg boy weighs himself in a lift descending with acceleration a. His apparent weight is only 600 N. Find a.

Solution

$$ma = mg - R \Rightarrow a = \frac{mg - R}{m} = \frac{70(9.81) - 600}{70} = \mathbf{1.23\ m\ s^{-2}}$$

6.5 Escape velocity

Escape velocity

is the **minimum velocity** a body needs at the surface of a planet to **overcome its gravitational pull and escape completely** into space.

Launch from the surface (A, r) with speed v; at infinity (B) the body only just arrives, so $U = 0$ and K.E. $\approx$ 0.

Conservation of energy: $\Sigma E_A = \Sigma E_B \Rightarrow -\frac{GMm}{r} + \frac{1}{2}mv^2 = 0 + 0$

$$\frac{1}{2}mv^2 = \frac{GMm}{r}$$

$$v_e = \sqrt{\frac{2GM}{r}} = \sqrt{2gr}$$

using $g = GM/r^2$; independent of the mass of the body

- From the Earth an object needs more than about **11.2 km s^{-1}** (40 320 km h^{-1}) to escape into space.
- Gas molecules in air (average speed about 0.5 km s^{-1} at s.t.p.; hydrogen about three times faster) are far slower than this, so the Earth keeps its atmosphere.
- A planet with a small escape velocity may lose its gases into space and have no atmosphere.

Example 6.5.1

(a) Find the escape velocity for a 5000 kg spacecraft launched from the Earth. (b) Determine its minimum kinetic energy to escape the Earth's field. (M_E = 5.98 × 10^{24} kg, R_E = 6.37 × 10^6 m)

Solution

(a) $v_e = \sqrt{\frac{2(6.67 \times 10^{-11})(5.98 \times 10^{24})}{6.37 \times 10^6}}$ = **1.12 × 10^4 m s^{-1}**

(b) K.E. = $\frac{1}{2}mv_e^2 = \frac{1}{2}(5.00 \times 10^3)(1.12 \times 10^4)^2$ = **3.14 × 10^{11} J**

Example 6.5.2

The Moon's radius is 1600 km and its surface field strength is 1.6 N kg^{-1}. Calculate the escape velocity from the Moon.

Solution

$$v_e = \sqrt{2gr} = \sqrt{2(1.6)(1.6 \times 10^6)} = \mathbf{2.3 \times 10^3\ m\ s^{-1}}$$

Example 6.5.3

The escape velocity from the Earth is 11.2 km s^{-1}. Find the escape velocity from a planet with twice the Earth's mass and three times its radius.

Solution

$$\frac{v}{v_E} = \sqrt{\left(\frac{M}{M_E}\right)\left(\frac{R_E}{r}\right)} = \sqrt{(2)(⅓)}$$

v = 11.2 × 0.816 = **9.15 km s^{-1}**

Exercises: escape velocity

1. Calculate the escape speed from the Earth for a 5000 kg spacecraft, and the kinetic energy it must have at the surface to move infinitely far away. (M = 6.0 × 10^{24} kg, R = 6.4 × 10^{6} m)

   Answer

   11.2 km s^{-1}; 3.14 × 10^{11} J

2. A planet has radius 5.0 × 10^{5} m and mean density 3.0 × 10^{3} kg m^{-3}. Calculate the escape velocity from its surface.

   Answer

   $M = \rho\left(\frac{4}{3}\pi R^3\right)$ = 1.57 × 10^{21} kg; $v = \sqrt{\frac{2GM}{R}}$ = 6.5 × 10^{2} m s^{-1}

Past-year STPM questions

STPM 2021 · Section C Q18(a) and (b)

(a) State Newton's law of universal gravitation. (b) Explain the relation between the high-tide phenomenon and the position of the Earth when it lies in line between the Moon and the Sun.

Solution

(a) Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of the masses of the particles and inversely proportional to the square of the distance between them.

(b) Tides are caused by the gravitational force of the Moon on the Earth's water. This force is inversely proportional to the square of the Earth–Moon distance, $F = Gm_E m_M / r_{EM}^2$. When the Sun, Earth and Moon lie in a straight line (new moon or full moon), the pulls of the Moon and the Sun act along the same line and add up, giving the highest high tides and the lowest low tides: **spring tides**.

Examiners' notes: (a) most candidates could state the law. Common mistakes: writing only "force" instead of gravitational (attractive) force; writing "radius between the two masses" instead of the distance between them. (b) Almost all candidates could not explain high tides and many left it blank. Not accepted: "closer" or "close" distance without relating force to distance; explaining with the Sun's gravity instead of the Moon's.

STPM 2021 · Section C Q18(c)

A satellite orbits the Earth in a circular path with a period of 6.2×10^3 s. Its mass is 5.5×10^3 kg. Determine (i) the radius of the orbit, (ii) the weight of the satellite, (iii) its potential energy.

Solution

(i) $\frac{GMm}{r^2} = \frac{4\pi^2 mr}{T^2} \Rightarrow r^3 = \frac{GMT^2}{4\pi^2} = 3.877 \times 10^{20} \Rightarrow \mathbf{r = 7.29 \times 10^6\ m}$

(ii) $W = \frac{GMm}{r^2} = \mathbf{4.12 \times 10^4\ N}$

(iii) $U = -\frac{GMm}{r} = \mathbf{-3.00 \times 10^{11}\ J}$

STPM 2021 · Section A Q7

Two planets A and B have masses M_A, M_B and radii R_A, R_B, with $M_A = ¼M_B$ and escape speeds $v_A = ⅓v_B$. What is R_A/R_B?

A. 0.19

B. 0.44

C. 2.25

D. 5.33

Answer

Answer: C

$$v = \sqrt{\frac{2GM}{R}} \Rightarrow R = \frac{2GM}{v^2}$$

$$\frac{R_A}{R_B} = \frac{M_A}{M_B}\left(\frac{v_B}{v_A}\right)^2 = ¼(3)^2 = \mathbf{2.25}$$

STPM 2023 · Section C Q19(a) and (b)

(a) State Newton's law of universal gravitation. [1] (b) Explain the change in gravitational acceleration when a meteoroid at a distance $3R$ from the surface of the Earth falls towards the surface, where R is the Earth's radius. [6]

Solution

(a) The gravitational force between two point masses is directly proportional to the product of their masses and inversely proportional to the square of the distance between them: $F = GM_1M_2/r^2$.

(b) Distance from the centre of the Earth: $r = R + 3R = 4R$, so $g = \frac{GM}{(4R)^2} = \frac{GM}{16R^2} = \frac{9.81}{16} = 0.61$ m s^{-2}.

As the meteoroid falls, r decreases and, since $g \propto 1/r^2$, g increases. At the surface $r = R$ and $g = 9.81$ m s^{-2}. The acceleration increases from 0.61 to 9.81 m s^{-2}, an increase of 9.20 m s^{-2}.

STPM 2023 · Section C Q19(c)

The Juno satellite is launched from the Earth to Jupiter and begins orbiting Jupiter at a radius of 7.05 × 10^4 km. The mass of Jupiter is 1.90 × 10^{27} kg. Determine (i) the escape speed of the satellite from the Earth's surface, [2] (ii) the distance travelled if it takes 1825 days to reach Jupiter, [2] (iii) the gravitational potential at its orbit, [2] (iv) the gravitational field strength at the satellite due to Jupiter. [2] (M_E = 5.97 × 10^{24} kg, R_E = 6.38 × 10^6 m, G = 6.67 × 10^{-11} N m^2 kg^{-2})

Solution

(i) $v = \sqrt{\dfrac{2GM_E}{R_E}}$ = **1.12 × 10^4 m s^{-1}**

(ii) t = 1825 × 24 × 3600 = 1.58 × 10^8 s; $s = vt$ = (1.12 × 10^4)(1.58 × 10^8) = **1.76 × 10^{12} m**

(iii) $V = -\dfrac{GM_J}{r}$ = **−1.80 × 10^9 J kg^{-1}**

(iv) $g = \dfrac{GM_J}{r^2}$ = **25.5 N kg^{-1}**